Esha Sharma
E-Commerce Funnel & Retention Analysis
Olist Brazil | 2016–2018 | 99,441 Orders

Tools Used:
SQL (MySQL), Python (pandas), Tableau Public

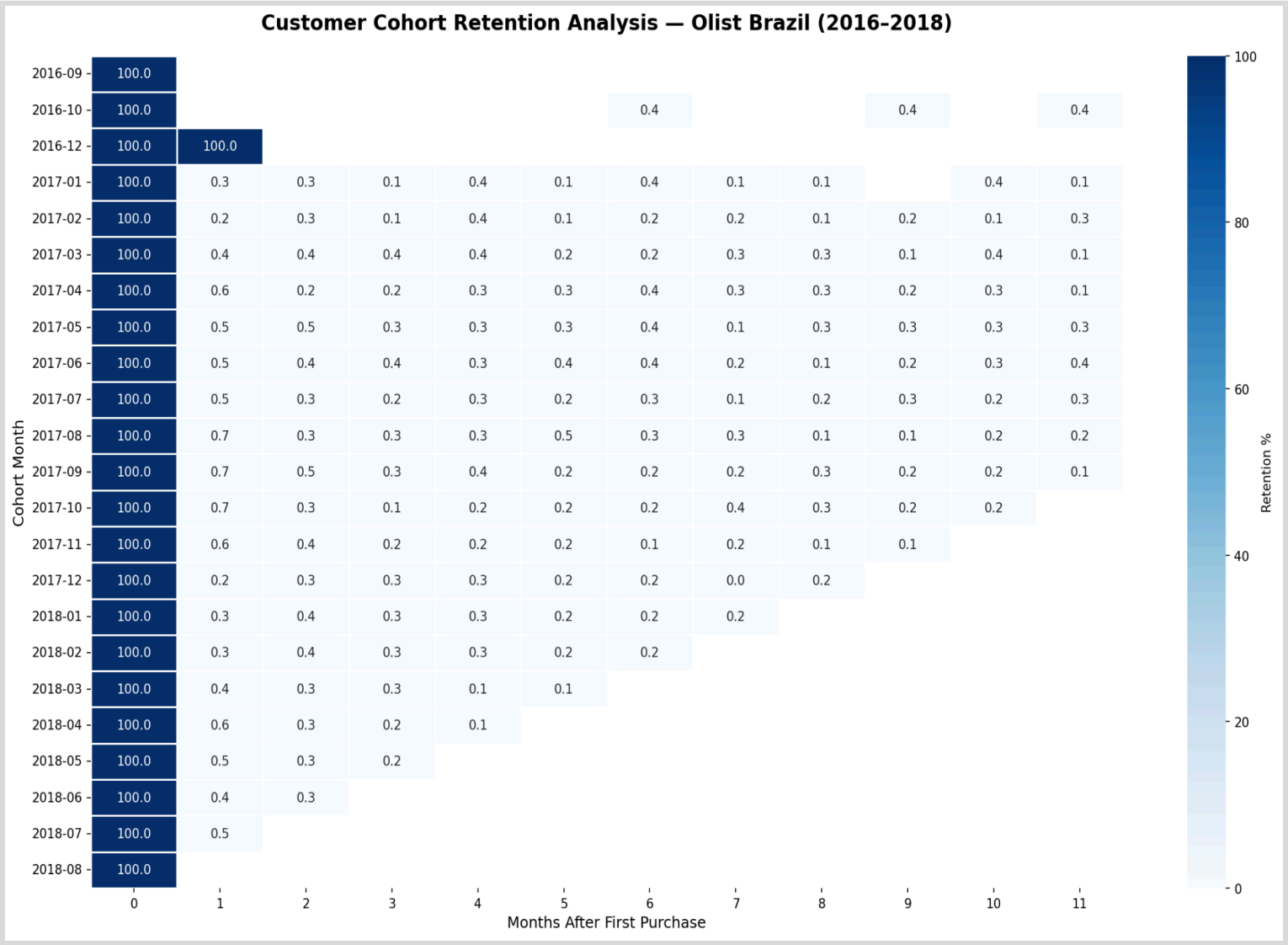
Aim:
To analyze customer behavior and identify drop-off points in the e-commerce funnel while evaluating retention trends.

Objective:
- Identify funnel drop-offs across key stages
- Analyze cohort retention patterns
- Recommend actionable business improvements

FINDING 1: SHIPPING IS THE CRITICAL DROP-OFF POINT
97,658 orders were shipped but only 96,476 delivered, a 1.79% drop-off at the final mile. At 99K order scale this = ~1,200 lost deliveries. Root cause: third-party logistics gaps in Brazil's regional distribution network.

FINDING 2 : REVENUE GROWTH STRONG BUT PLATEAUING
Monthly revenue grew from ~R$50K (Oct 2016) to R$1.2M peak (Nov 2017), a 24x increase in 14 months. However revenue plateaued through 2018 suggesting customer acquisition is slowing and retention must compensate.

FINDING 3 : MONTH-2 RETENTION ONLY 30–35% ACROSS ALL COHORTS
Month-2 retention is ~30–35% across all cohorts. The business is heavily dependent on new customer acquisition. A 10% improvement in month-2 retention would add ~R$170K monthly recurring revenue.


Customer Cohort Retention Analysis — Olist Brazil (2016–2018)
Cohort Month
Months After First Purchase
Retention %
2016-09 100.0
2016-10 100.0 0.4 0.4 0.4
2016-12 100.0 100.0
2017-01 100.0 0.3 0.3 0.1 0.4 0.1 0.4 0.1 0.1 0.4 0.1
2017-02 100.0 0.2 0.3 0.1 0.4 0.1 0.2 0.2 0.1 0.2 0.1 0.3
2017-03 100.0 0.4 0.4 0.4 0.4 0.2 0.2 0.3 0.3 0.1 0.4 0.1
2017-04 100.0 0.6 0.2 0.2 0.3 0.3 0.4 0.3 0.3 0.2 0.3 0.1
2017-05 100.0 0.5 0.5 0.3 0.3 0.3 0.4 0.1 0.3 0.3 0.3 0.3
2017-06 100.0 0.5 0.4 0.4 0.3 0.4 0.4 0.2 0.1 0.2 0.3 0.4
2017-07 100.0 0.5 0.3 0.2 0.3 0.2 0.3 0.1 0.2 0.3 0.2 0.3
2017-08 100.0 0.7 0.3 0.3 0.3 0.5 0.3 0.3 0.1 0.1 0.2 0.2
2017-09 100.0 0.7 0.5 0.3 0.4 0.2 0.2 0.2 0.3 0.2 0.2 0.1
2017-10 100.0 0.7 0.3 0.1 0.2 0.2 0.2 0.4 0.3 0.2 0.2
2017-11 100.0 0.6 0.4 0.2 0.2 0.2 0.1 0.2 0.1 0.1
2017-12 100.0 0.2 0.3 0.3 0.3 0.2 0.2 0.0 0.2
2018-01 100.0 0.3 0.4 0.3 0.3 0.2 0.2 0.2
2018-02 100.0 0.3 0.4 0.3 0.3 0.2 0.2
2018-03 100.0 0.4 0.3 0.3 0.1 0.1
2018-04 100.0 0.6 0.3 0.2 0.1
2018-05 100.0 0.5 0.3 0.2
2018-06 100.0 0.4 0.3
2018-07 100.0 0.5
2018-08 100.0
0 1 2 3 4 5 6 7 8 9 10 11
100 80 60 40 20 0

Recommendations & Prioritization

RECOMMENDATION 1 : Launch Month-2 Retention Email Campaign

- **What:** Automated email at day-25 post-purchase with personalized reorder suggestions based on category.
- **Impact:** HIGH — directly addresses 65% month-2 drop-off
- **Effort:** LOW — email automation, no engineering needed
- **Expected outcome:** +8–12% month-2 retention

RECOMMENDATION 2 : Logistics SLA Audit for Final-Mile Delivery

- **What:** Identify which seller regions have highest non-delivery rate, set SLA thresholds, switch underperforming carriers
- **Impact:** HIGH — recovers ~1,200 lost deliveries/month at full scale
- **Effort:** HIGH — requires carrier contract renegotiation
- **Expected outcome:** Delivery rate 97% → 99%

RECOMMENDATION 3 : Double Down on Bed & Bath + Health & Beauty

- **What:** Increase seller recruitment and paid acquisition budget for top 2 revenue categories
- **Impact:** MEDIUM — already performing, incremental upside
- **Effort:** LOW — budget reallocation, no new systems
- **Expected outcome:** +15–20% revenue in top 2 categories in 6 months

❖ Impact vs Effort Matrix:

Impact	Effort	Items
High	High	R2 – Delivery Audit (Strategic Investment)
High	Low	R1 – Email Optimization
Low	Low	R3 – Category Focus (Quick Wins 🔥)

Interactive Dashboard (Tableau Public):
https://public.tableau.com/views/OlistEcommerceFunnelAnalysisEshaSharma_17761702456240/Dashboard1?:language=en-US&:sid=&:display_count=n&:origin=viz_share_link

GitHub Repository:
https://github.com/promptedbyesha/Olist-Ecommerce-Analysis

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Olist Brazil (2016–2018) | April 2026